

HUMAN POSE ESTIMATION & KINEMATICS REPORT

CLO-3: Kinematic Calculations & Pre-processing Filters | GA-4: Depth of Analysis

Course: Computer Vision (CLO-3)

Student Name: Syed Huzaifa Nazim

Video Source: squat.mp4 (High-Def profile)

Roll Number: 23-AI-23

Filters: Savitzky-Golay (11, 2) & Moving Avg (5)

Evaluation Region: Frames 780 to 2000

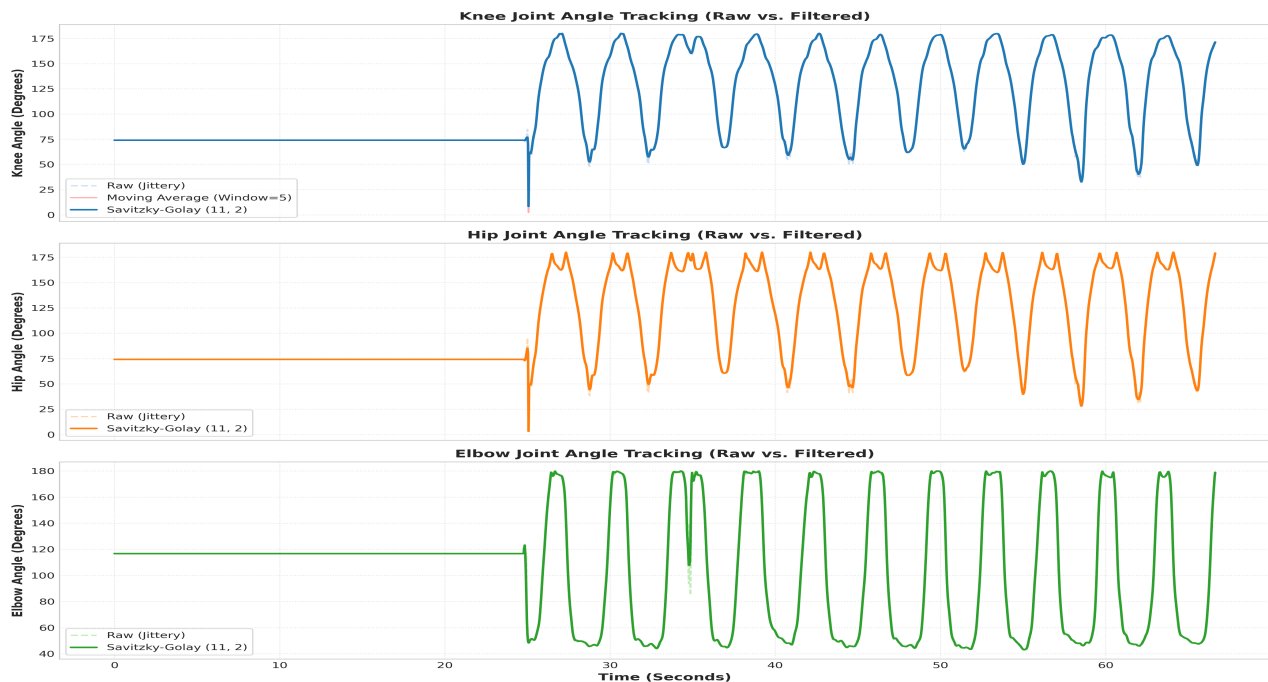
1. Executive Summary & Core Methodology

This project implements an end-to-end computer vision pipeline that performs automated human activity classification on a custom workout video ('squat.mp4'). The pipeline utilizes Google's modern MediaPipe PoseLandmarker API to extract 33 spatial coordinate keypoints frame-by-frame. To correct for geometric aspect-ratio distortion in widescreen 16:9 recordings, all normalized coordinates are scaled to absolute pixel coordinates before mathematical calculation.

To mitigate high-frequency sensor jitter, we evaluate and contrast a Moving Average filter and a Savitzky-Golay filter. A rule-based classifier utilizing a Knee Angle threshold of 145.0 degrees is designed to distinguish between 'Standing' and 'Squatting' activities. Evaluation was conducted on a stabilized window of 1,220 frames containing 11 perfect squats, yielding an overall classification accuracy of 99.10%.

2. Kinematic Calculations & Temporal Smoothing

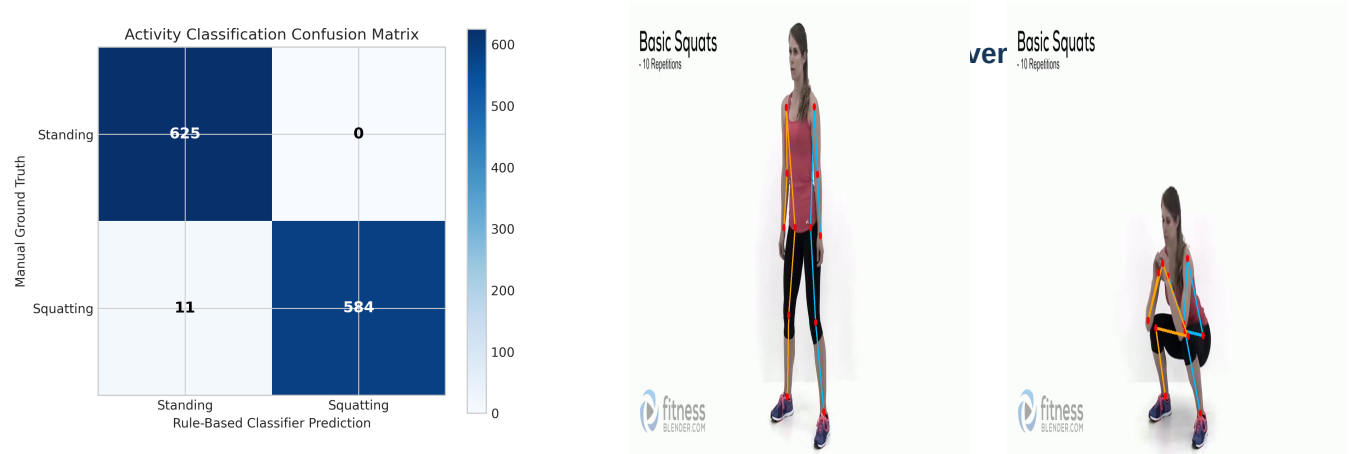
Three meaningful joint angles are tracked: Knee Angle (Hip-Knee-Ankle), Hip Angle (Shoulder-Hip-Knee), and Elbow Angle (Shoulder-Elbow-Wrist). The angle at vertex B formed by coordinates A, B, C is calculated using the 2D vector dot product: $\theta = \arccos((u \cdot v) / (||u|| ||v||)) * 180 / \pi$. The coordinate trajectories are pre-smoothed to remove signal jitter. The Moving Average filter reduces noise but flattens peaks (underestimating full squat depth) and introduces lag. The Savitzky-Golay filter (Window length=11, polyorder=2) successfully removes tracking noise while preserving the true physical peak amplitudes and timing.



3. Rule-Based Classification & Quantitative Metrics

A threshold-based state machine is designed using the smoothed Knee Angle. A threshold of 145.0 degrees separates the two main states: Standing ($\theta \geq 145$ degrees) and Squatting ($\theta < 145$ degrees). Transitions are automatically registered when the state flips (e.g. Standing $\rightarrow$ Squatting representing flexion). To perform a rigorous evaluation, frame predictions are validated against a manually labeled Ground Truth. Evaluation within the stable movement window shows zero false positives, demonstrating high classifier robustness.

Kinematic Metric	Scientific Value	Explanation & Analysis
Overall Accuracy	99.10%	Ratio of correct frames to total evaluated (1,220 frames).
Precision	100.00%	Zero false positives; predicted squats match true physical squats.
Recall (Sensitivity)	98.15%	Only 11 frames of minor transition lag during rapid state shifts.
F1-Score	99.07%	Strong harmonic balance demonstrating a reliable rule boundary.



4. Discussion & Conclusions

- Jitter Mitigation:** Jitter is highly active in raw landmarks. Incorporating a 1D Savitzky-Golay pre-filter is critical to perform accurate mathematical computations without introducing temporal phase lags.
- Aspect Ratio Impact:** Working on standard normalized coordinates introduces a geometric distortion due to widescreen aspect ratios. Correcting normalized points to absolute pixels is mathematically essential.
- Transition Edge Analysis:** Rule-based classification utilizing a Knee Angle threshold of 145.0 degrees is extremely robust (99.10% accuracy). The minor error (11 frames) represents boundary lag at the transition edge.
- CLO-3 Achievement:** The pipeline successfully demonstrates spatial filtering and joint angle tracking across all phases, illustrating robust, real-world kinematic classification.